

A Conceptual Framework for Chronic Kidney Disease Self-Management Support

Abstract

Chronic kidney disease (CKD) is increasingly becoming a major challenge in long-term healthcare because it requires continuous monitoring, lifestyle management, medication adherence, and communication with healthcare professionals beyond formal clinical consultations. Existing digital-health tools for CKD often focus on isolated functions such as education, symptom reporting, reminders, dietary logging, or patient support. In this context, this paper proposes a conceptual framework for clinically responsible CKD self-management. This study aims to develop a structured approach to integrating patient-generated health data (PGHD), clinical information, guideline-informed knowledge, and clinical review into a single decision-support pathway. The framework was developed through a synthesis of relevant literature, identified gaps, and self-management requirements reported in CKD digital health, PGHD, and clinical decision support. The proposed concept consists of three connected layers: a data layer, a rule-based analysis layer, and a clinician review layer. This paper presents a preliminary attempt to organise a supported CKD self-management as decision support rather than autonomous clinical decision-making, providing a basis for future expert review, prototype refinement, and empirical validation.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Human-centered computing** → *Interactive systems and tools*; • **Information systems** → *Decision support systems*.

Keywords

Chronic Kidney Disease, Digital Health, CKD Self-Management, Health-Tech, Clinical Decision Support, Patient-Generated Health Data

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1 Introduction

Chronic kidney disease (CKD) is increasingly becoming a major public-health and clinical-management challenge. CKD is a long-term condition in which kidney structure or function is persistently abnormal, commonly indicated by reduced kidney filtration, albuminuria, or other markers of kidney damage lasting for more than three months [25]. It is progressive, frequently under-recognised, and associated with substantial morbidity and mortality. CKD is responsible for

long-term clinical monitoring needs, cardiovascular risk, and health-service use. There is increasing pressure on patients to manage complex treatment requirements in everyday life [25]. This makes early detection, continuous self-management support, and clinically meaningful monitoring critical to reducing avoidable deterioration and improving patient-centred care.

Traditionally, CKD management has been guided by clinical assessment, laboratory indicators, guideline-based monitoring, and patient education. In this setting, healthcare professionals assess kidney function, symptoms, comorbidities, blood pressure, medication use, and lifestyle-related risk factors to guide treatment and follow-up [25]. However, many aspects of CKD management occur outside clinical encounters, where patients are required to follow treatment advice, monitor symptoms, and manage lifestyle-related behaviours in their daily lives. Therefore, self-management support has become a critical part of CKD care, as patients are expected to manage medication adherence, dietary restrictions, fluid intake, symptoms, appointments, sleep, mood, and communication with clinicians outside formal consultations [7, 28]. CKD self-management can be defined as the daily process through which patients monitor and manage treatment requirements and lifestyle behaviours in collaboration with healthcare professionals [19]. Patient-generated health data (PGHD) refers to health-related information recorded by patients outside clinical encounters, including symptoms, medication use, food and fluid intake, mood, sleep, physical activity, and home-monitoring data.

Although current clinical and educational approaches are necessary, they suffer from important limitations. First, self-management information is often subjective, recorded irregularly, and difficult to interpret across multiple domains such as symptoms, diet, medication, fluid intake, sleep, and mood [20, 21]. Second, much of this information remains confined to clinical encounters, meaning that clinicians may have limited visibility into patients' day-to-day condition between appointments [5]. Third, clinical decision-support systems may generate alerts or recommendations, but their value remains limited when outputs are not actively reviewed, contextualised, or followed up by healthcare professionals [29]. The central limitation is therefore not merely the availability of patient data, but the absence of a structured mechanism for turning this data into clinically accountable feedback.

Existing solutions have not yet realised this potential in a clinically integrated way. Previous work has focused only on isolated functions, including patient support, education, activation, dietary logging, reminders, communication, and selected self-care behaviours [12, 13]. The reviewed literature suggests that an important gap in CKD self-management is the limited integration of patient-generated health data, clinician-facing interpretation, and human review within a

single decision-support pathway [17, 26]. Similarly, although patient-generated health data and electronic symptom monitoring may improve visibility between consultations, it has not yet been established how these data should be converted into clinically meaningful and reviewable insights [18, 27].

In response to these challenges, this paper proposes a conceptual framework for self-management of CKD. The framework has not yet been empirically validated and is therefore presented as a proposed structure rather than as a tested intervention. It is used to examine how patient-generated health data and clinician review might be organised within a clinically responsible decision-support pathway. Its design is informed by three requirements identified in the reviewed literature: connecting fragmented mobile self-management functions, interpreting patient-generated health data in a clinically meaningful way, and maintaining human accountability.

The research question under investigation is: How can a CKD self-management framework integrate patient-generated health data and clinician review to support patient care?

The conceptual framework consists of three connected layers. The first layer captures multi-domain PGHD and real-world data from patients' day-to-day lives, including symptoms, medication adherence, food and fluid intake, sleep, mood, activity, appointments, and other aspects of daily functioning, alongside a personalised guideline-informed knowledge base with clinician-defined targets. The second layer applies rule-based analysis to organise longitudinal self-management data into patterns, trends, and potential review signals. The third layer involves clinical review, where healthcare professionals interpret, validate, and contextualise these outputs before feedback or action is communicated to the patient.

The conceptual framework emphasises integration, interpretability, and may support more frequent and meaningful monitoring across both home and clinical settings. In doing so, the framework responds to the need for CKD self-management systems that can support patients' daily care while preserving professional judgement and clinical safety.

2 Background and Related Work

There is a considerable body of literature establishing chronic kidney disease (CKD) as an increasing global health burden. The Global Burden of Disease Study (GBD) CKD Collaborators (2025) estimated that 788 million adults had CKD in 2023, compared with 378 million in 1990 [4]. Earlier, the GBD CKD Collaboration (2020) reported 697.5 million cases in 2017 and a 41.5% increase in the all-age CKD mortality rate between 1990 and 2017 [3]. KDIGO (2024) further defines monitoring of kidney function, blood pressure, medication use, symptoms, diet, fluid intake, and modifiable risk factors as central to CKD management [25].

Numerous studies have proposed that CKD self-management encompasses medication, symptoms, diet, fluid, sleep, and emotional health by examining patients' work outside clinical settings. But these studies have tended to focus on haemodialysis and English-speaking populations, limiting applicability

to earlier-stage CKD and diverse groups [9, 20]. Potential research, such as Flythe et al. (2018), found fatigue, cramping, and body aches in 94%, 79%, and 76% of surveyed haemodialysis patients [2]. Roberti et al. (2018) describe CKD self-management as time-consuming, invasive, and exhausting for patients and caregivers [21].

In recent years, there has been growing interest in digital and mobile interventions for CKD self-management. Various approaches have been proposed to support dietary logging, education, reminders, activity monitoring, and patient-provider communication [12, 14, 15]. Li et al. (2020) conducted a randomised study in which the intervention group achieved higher self-efficacy scores than controls, 171.28 versus 142.21, higher self-management scores, 54.16 versus 47.58 and a smaller eGFR decline of -0.56 versus -4.58 mL/min/1.73 m² [13]. However, this approach is limited by a small, single-hospital sample, a short 90-day follow-up, and the inability to separate the effects of different intervention components.

A potential research review found that the available evidence was of low quality and too heterogeneous for meaningful summary estimates [26]. Other studies report potential improvements in activation and communication, but a key problem with much of the literature on mobile CKD care is that applications address individual functions, while integrated interpretation of multiple self-management variables remains unclear [17, 23, 24].

Recent studies have explored patient-generated health data (PGHD), wearable devices, and electronic patient-reported outcome measures. For example, Evans et al. (2020) found that seven of eight sites achieved symptom-screening rates above 80%, suggesting that routine electronic screening is feasible [1]. Similarly, Vieceilli et al. (2022) reported that patients, nurses, and nephrologists considered electronic symptom monitoring with clinician feedback to be feasible, provided there was clinician engagement, reliable technology, and timely responses [27].

Clinical decision support has been widely investigated as a mechanism for organising CKD information and providing guideline-linked recommendations. Gulla et al. (2017) identified 21 requirements for a CKD decision-support tool, including actionable information, guideline links, and trend visualisation [6]. Litvin et al. (2016) reported a median 30% increase in albuminuria screening, academic detailing and performance reports prevented isolation of the decision-support effect [16]. A serious limitation of automated decision support is that alerts may be generated without active follow-up, contextual interpretation, or clear accountability. The main challenge is not only identifying risk but also clarifying who is responsible for reviewing outputs, assessing urgency, and monitoring follow-up actions.

3 Research Gap and Motivation

Despite substantial work on Digital self-management, CKD prediction, and clinical decision support, three interconnected deficits remain.

First, current solutions to CKD self-management remain functionally fragmented. In recent years, there has been growing interest in applications such as My Kidneys & Me (MK&M), a co-designed digital program providing tailored education and self-management support, which was associated with increased patient activation among engaged users, yet 47% of outcome data were missing at follow-up. Moreover, the cohort consisted predominantly of a digitally capable Caucasian population, which raises concerns regarding generalisability [14]. Previous work has only focused on education, communication, activation, or selected self-care behaviours and has failed to address how symptoms, medication adherence, fluid intake, diet, sleep, emotions, and appointments can be interpreted together as clinically meaningful information.

Second, research has tended to focus on CKD detection, progression prediction, mortality estimation, and risk stratification rather than daily self-management [10]. Kanda et al. (2023) demonstrated strong predictive performance using random-forest models; however, their data did not sufficiently include lifestyle, food intake, family history, or social determinants [8]. This distinction can be represented conceptually as:

$$\hat{R} = f(X_{\text{clinical}})$$

where $\hat{R}$ denotes the predicted CKD risk using X_{clinical} , clinical and laboratory data.

Motivated by these limitations, this study argues that the central issue is not the absence of digital tools or predictive models, but the absence of a clinically responsible pathway that connects patient-generated self-management data with guideline-informed interpretation, analysis, and human review. Existing mobile CKD applications are capable of capturing patient self-management data; however, the utility of such data remains constrained when it is not systematically integrated across multiple domains and translated into clinically actionable insights. Likewise, although prediction-oriented ML models have demonstrated effectiveness in estimating CKD risk using clinical and laboratory variables [22], they provide limited support for understanding and managing the day-to-day behaviours and experiences that influence disease progression, including medication adherence, dietary practices, fluid management, symptom monitoring, sleep patterns, psychological well-being, and appointment adherence.

The motivation for this study is therefore to formulate a framework that shifts CKD support from isolated prediction toward integrated self-management assistance. In this context, a rule-based model is being proposed as an analytical layer that can identify trends, repeated patterns, and clinically relevant combinations in longitudinal patient-generated data. Clinical review is included to preserve clinical judgement, accountability, and patient safety. This study is therefore motivated by the need for a framework that can organise fragmented self-management data, support interpretable insight generation, and provide a structured pathway from patient records to clinician-reviewed feedback.

4 Methodology

This study aimed to develop a conceptual framework for chronic kidney disease (CKD) self-management that addresses limitations identified in existing mobile health applications and clinical decision-support research. In order to identify the central challenges associated with CKD self-management, we reviewed current literature, analysed existing approaches, and examined the reported limitations of mobile applications, patient-generated health data (PGHD) and clinical decision-support systems. Previous work has only focused on isolated aspects of CKD management, whereas this study seeks to integrate these components within a single framework. Consequently, the framework was conceptualised through a synthesis of existing evidence, identified research gaps, and domain-specific requirements reported in the literature.

To enable us to investigate current approaches to CKD self-management and the use of patient-generated health data, information was obtained from peer-reviewed journal articles, review articles, case studies, recent and credible studies published in recognised journals and conferences and theoretical models identified during the literature review process. Particular attention was paid to the broader value of structured PGHD beyond immediate patient support. The framework is intended initially to help patients record and manage daily information related to CKD. Systematic collection of real-world PGHD can also allow future analysis of symptoms, trends, functional changes, adherence patterns and other factors associated with CKD. This is important because clinically meaningful markers and patient-reported indicators in CKD remain complex and contested. Therefore, the proposed framework is positioned not only as a self-management support model but also as a structure for generating data that may inform future CKD monitoring, research, and care refinement. The conceptual framework is composed of three interconnected layers:

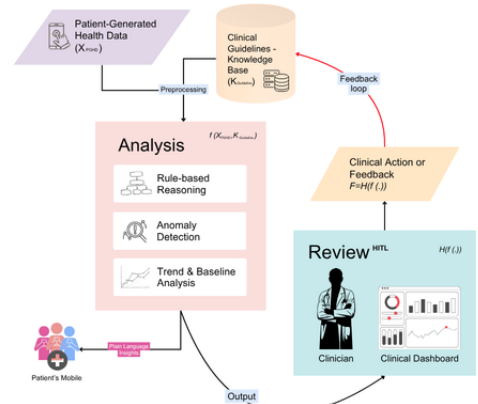


Figure 1: Proposed CKD self-management framework integrating data input, analysis and clinical review.

- **Data Input Layer** (Processed): Aggregates patients' information through patient-generated health data and personalised guideline-based knowledge.
- **Analysis Layer** (Rule): Employs a rule-based system approach combining rule-based reasoning, trend and baseline analysis and anomaly detection.
- **Review Layer** (HITL): Incorporates a mechanism whereby clinicians review, validate, and contextualise generated outputs to patients.

4.1 Data Input Layer

The Data Layer provides the input foundation of the concept. It brings together two types of information:

$$X = [X_{\text{PGHD}}, K_{\text{guideline}}]$$

X_{PGHD} means patient-generated health data collected through the patient-facing front end, such as a mobile application used for daily self-management. Patients will regularly enter information related to their day-to-day health and activities, including symptoms, medication adherence, food intake, fluid intake, sleep, emotions, physical activity, dialysis reminders, and GP or specialist appointments. The mobile application will support continuous data collection through routine patient use and notification prompts, ensuring that longitudinal self-management data are captured and stored for subsequent analysis within the framework. Table 1 describes the proposed mobile application features used to collect patient-generated health data for CKD self-management.

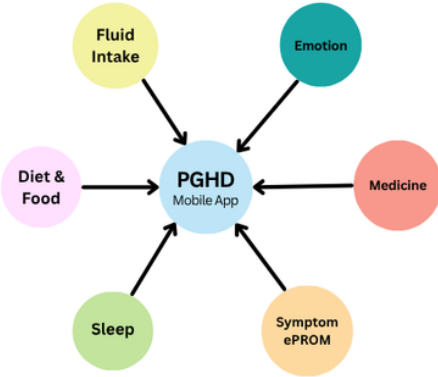


Figure 2: Patient-generated health data flow within the proposed CKD self-management framework.

$K_{\text{guideline}}$ refers to the personalised knowledge base that supports the interpretation of patient data within the framework. This knowledge base includes data such as CKD stage and estimated glomerular filtration rate (eGFR) and incorporates relevant clinical guidelines, including KDIGO guidelines, clinician-defined targets, patient-specific rules, CKD management recommendations, and feedback provided by clinicians

during each patient visit. By combining standardised clinical knowledge with individualised recommendations, $K_{\text{guideline}}$ helps ensure that patient data are analysed within an appropriate clinical and personal context.

The purpose of this layer is to make sure that the analysis does not depend only on isolated patient entries. Instead, patient records are interpreted together with clinical context and personalised rules.

4.2 Analysis Layer

The second layer of the conceptual framework is the Analysis Layer. The purpose of this layer is to transform heterogeneous patient-generated data into structured analytical outputs that can subsequently be reviewed. In order to analyse the self-management data for CKD in a clinically meaningful way, this layer does not examine patient entries in isolation. Instead, it integrates patient-generated health data and personalised guideline-based knowledge.

The design of this layer was based on the need to interpret self-management data in relation to the clinical context. For example, a high fluid-intake entry has limited meaning when viewed in isolation. However, when analysed alongside the patient's fluid target, CKD stage, swelling history, and clinician-defined rules, it becomes more clinically meaningful. Therefore, the Analysis Layer uses X_{PGHD} to provide longitudinal self-management records and $K_{\text{guideline}}$ to provide the rules and thresholds for interpretation.

$$\text{Output} = f(X)$$

The output may include repeated pattern signals, anomaly flags, trend summaries, alerts, and patient-facing self-management insights. In this layer, the function $f(X)$ refers to a rule-based model process consisting of rule-based reasoning, trend and baseline analysis and anomaly detection.

4.2.1 Rule-Based Reasoning: The first component of the Analysis Layer is rule-based reasoning that applies predefined rules and thresholds to patient data. This method was chosen because it is practical, transparent and suitable for an early-stage clinical framework where large labelled datasets may not yet be available.

$$R = f_r(X)$$

For example, if a patient repeatedly records missed medication, the system can compare this behaviour with the medication-adherence rule stored in $K_{\text{guideline}}$. If the patient also has uncontrolled blood pressure or advanced CKD, the same missed-medication pattern may be assigned greater importance. The resulting output may be an adherence-related concern or a review flag. This shows that rule-based reasoning is not simply checking whether a value exists; it interprets the patient's record against personalised rules and clinical context. For instance:

$$\text{Missed medication} \geq 3 \text{ times in 7 days} \rightarrow R_{\text{adherence}}$$

where $R_{\text{adherence}}$ represents a concern about medication-adherence. Similarly, if a patient reports worsening fatigue,

Table 1: Proposed Mobile Application Features for X_{PGHD}

Parameter	Description	Impact
Symptom Tracking	Allows patients to record CKD-related symptoms such as fatigue, swelling, breathlessness, pain, nausea, and itching.	Enables early identification of symptom deterioration and supports trend analysis over time.
Medication Adherence Monitoring	Allows patients to record medication intake and receive reminders for prescribed treatments.	Improves medication adherence, identifies missed doses, and supports clinician review of medication-related issues.
Fluid Intake Tracking	Enables patients to log daily fluid consumption against personalised targets.	Supports fluid management, helps detect deviations from prescribed limits, and reduces the risk of complications.
Dietary Monitoring	Allows patients to record meals, nutritional intake, and CKD-specific dietary restrictions.	Encourages healthier dietary behaviours and supports personalised dietary feedback.
Sleep Monitoring	Enables patients to record sleep duration and quality.	Provides insights into wellbeing and identifies patterns associated with worsening symptoms or poor health outcomes.
Mood and Emotional Wellbeing Tracking	Allows patients to record mood, stress, anxiety, and emotional concerns.	Supports holistic self-management and helps identify psychosocial factors affecting health.
Physical Activity Tracking	Enables recording of exercise and daily activity levels.	Encourages healthy lifestyle behaviours and supports monitoring of functional status.
Appointment Management	Stores and provides reminders for GP, specialist, dialysis, and laboratory appointments.	Improves attendance and supports continuity of care.
Dialysis Reminders and Logging	Allows patients undergoing dialysis to record dialysis sessions and receive reminders.	Supports treatment adherence and provides additional context for clinical review.
Personalised Notifications	Sends reminders for data entry, medications, appointments, and self-management activities.	Encourages sustained engagement and improves the completeness of patient-generated health data.
Trend Visualisation Dashboard	Displays graphs and summaries of symptoms, medication adherence, fluid intake, and other tracked variables.	Helps patients understand health patterns and promotes self-awareness.
Rule-Based Alerts	Generates alerts when predefined clinical thresholds or clinician-defined rules are exceeded.	Provides timely identification of potential concerns and supports proactive intervention.
Feedback Messages	Provides understandable insights for alerts and recommendations generated by the system.	Improves transparency, trust, and patient understanding of system outputs.

reduced activity, and repeated missed appointments, the system can compare these observations with the relevant rules and thresholds stored in $K_{\text{guideline}}$. When the recorded pattern satisfies a predefined condition, the system may generate a self-management concern or review flag.

4.2.2 Trend and Baseline Analysis: The second component is trend and baseline analysis. In order to identify changes over time, this component is a time-based analytical method that compares current patient records with previous records or the patient’s own baseline. This is necessary because CKD self-management is longitudinal. A single data entry may not be meaningful, whereas repeated changes over several days or weeks may indicate a pattern requiring review.

$$T = f_t(X_{\text{PGHD}}^t, B_{\text{patient}})$$

where (T) denotes the trend output, X_{PGHD}^t denotes the current patient-generated data at time (t), and B_{patient} denotes the patient’s baseline. The patient’s baseline can be understood as the usual pattern of the patient’s previous

records. The trend can then be described as:

$$T = X_{\text{current}} - B_{\text{patient}}$$

For example, if a patient usually reports a fatigue score of 3 out of 10 but reports scores of 6 or 7 over several days, the system identifies a worsening fatigue trend. If this trend occurs together with poor sleep, reduced activity, or missed medication, the system can generate a broader self-management trend summary. This does not mean that the system makes a clinical judgement. It means that the system identifies a change from the patient’s usual pattern and prepares this information for explanation and review. Trend analysis may be applied to several self-management domains, including symptom severity, medication adherence, sleep quality, mood score, fluid intake, dietary behaviour, activity level, and appointment attendance.

4.2.3 Anomaly Detection: The third component is anomaly detection. This component identifies unusual changes or unexpected combinations in the patient’s data. It differs from rule-based reasoning because it does not only check whether a predefined rule has been triggered. Instead, it determines

whether the current patient pattern is different from the patient's usual behaviour.

$$A = f_a(X_{\text{current}}, B_{\text{patient}})$$

where (A) denotes the anomaly output as in the difference between the current pattern and the usual pattern, X_{current} denotes the current patient pattern, and B_{patient} denotes the patient's normal or expected pattern.

For example, a patient can usually demonstrate stable adherence to medications, consistent sleep patterns, regular attendance at appointments, and moderate levels of activity. If the same patient suddenly records repeated missed medication doses, poor sleep quality, reduced activity, low mood, and missed appointments over several days, the anomaly-detection component may identify this combination as an unusual self-management pattern when compared with the patient's established baseline. This output does not indicate a diagnosis. It indicates that the current pattern differs from the patient's previous behaviour and may need clinician review.

A simple anomaly-detection approach can be used at the initial stage by comparing current patient-generated health data with the patient's baseline behaviour and predefined personalised thresholds. An anomaly is identified when the current pattern deviates substantially from the patient's usual pattern or exceeds a clinically relevant threshold defined within the personalised knowledge base. This approach allows the system to detect unusual self-management behaviours without requiring labelled training data. However, considerable attention must be paid when applying these techniques in a clinical setting, because unusual does not automatically mean clinically serious.

For this reason, anomaly outputs must be passed to the review layers as a data dashboard to the clinicians, as clinicians require sufficient information to review and contextualise the output. Also, plain language insights are being sent out to the patient's mobile application, explaining clear and non-alarming insights that help them understand their own data. As an example, Patient-facing insights might say: "Your recent records show several missed medication entries this week. This may be important for your care plan and may be reviewed by your clinician."

4.3 Review Layer

The third layer of the conceptual framework is the Clinician Review Layer. The purpose of this layer is to ensure that outputs generated by the Analysis Layer are reviewed, interpreted, and contextualised by clinicians before clinical action or feedback is provided. This layer was included because rule-based outputs may identify patterns, alerts, anomalies, or signals, but these outputs still require professional interpretation in relation to the patient's broader clinical condition.

$$F = H(O, f(X))$$

The Review Layer receives data through a clinical dashboard, which serves as the main interface through which clinicians view and assess patient information, and receives the

analytical output from the Analysis Layer. This output may include trends, alerts, anomaly flags and self-management summaries. The clinician can then review the output, validate whether it is clinically relevant, contextualise it using the patient's clinical history, current status, and previous records, before deciding whether any feedback or action is appropriate. The feedback may include no action, continued monitoring, patient feedback, follow-up communication, adjustment of personalised targets, or further clinical assessment.

Once the clinician has reviewed the output, the resulting feedback is returned to the personalised knowledge base and data store. This feedback loop is important because it allows the system to retain clinician input for future analysis. For example, if a clinician adjusts a patient's fluid target, updates a symptom-monitoring rule, or records that a particular alert did not require action, this information can be stored and used when future patient records are analysed.

$$K_{\text{guideline}}^{t+1} = K_{\text{guideline}}^t + F_{\text{clinician}}$$

This does not mean that the system changes clinical guidelines themselves. Rather, it updates the patient-specific data store, including clinician feedback, personalised targets, review history, and rules used for future interpretation. This ensures that future analysis is informed by previous clinician review rather than relying only on raw patient-entered data.

As this study proposes a conceptual framework, empirical validation falls outside the scope of the present work and is identified as future research. To find out the framework's suitability for practical deployment, future studies may employ expert evaluation, gathering proper past clinical data, pilot implementation, and case-based assessment. Validation criteria include usability, interpretability, clinical relevance, transparency, workflow integration, and stakeholder acceptance. In addition, clinician feedback and patient perspectives could be collected to assess whether the framework provides meaningful support for CKD self-management.

5 Discussion

This study has presented a conceptual framework for the self-management of chronic kidney disease that is intended to support, rather than replace, clinical judgement. A key contribution of the concept is its focus on supporting CKD self-management beyond the clinical consultation. CKD requires patients to manage multiple daily activities; however, these are often addressed separately. The concept addresses this issue by connecting multiple self-management domains within one structured pathway. This would allow patient-generated health data to be interpreted not only as isolated entries but as longitudinal patterns that may indicate changes in the patient's self-management condition.

The framework also has important implications for improving clinical visibility between consultations. In usual care, clinicians may only observe a patient's condition during scheduled appointments or through periodic laboratory results. This can limit awareness of daily symptom changes,

adherence difficulties, lifestyle patterns, or emerging self-management concerns. By collecting and analysing patient-generated health data over time, the proposed system can generate alerts, trends, and anomaly flags. These outputs may provide clinicians with a more continuous view of the patient's condition.

This may also contribute to safety and trust by ensuring that patient-facing outputs are presented as self-management insights rather than clinical judgements, while clinician-facing outputs include sufficient data context for review. In addition, the feedback loop from clinician review to the personalised knowledge base allows future analysis to be informed by previous clinical judgement; however, these implications should be interpreted with caution, as the framework does not modify formal clinical guidelines, which remain external clinical references.

Taken together, existing studies have largely focused on isolated functions such as education, dietary logging, symptom reporting, or prediction. In contrast, this framework brings these components together within a layered structure and highlights the potential value of structured PGHD to be collected over time for further clinical analysis. Such data may help identify patterns in symptom burden, adherence challenges, patient behaviour, and clinician feedback, thereby supporting future refinement of CKD self-management interventions. However, realising this model would require careful consideration of privacy, security, data governance, and clinical integration. Particular attention would also be needed if AI or generative AI tools were incorporated in future versions, as inappropriate integration could increase the risk of health data leakage or unsafe use of sensitive patient information.

6 Conclusion

In conclusion, this paper has presented a conceptual framework for CKD self-management that suggests a structured way to support self-management beyond the clinical consultation by transforming daily patient records into trends, alerts, anomaly signals and plain language insights. The value of the proposed framework lies in its layered design, where clinicians remain responsible for reviewing outputs and determining appropriate feedback or action.

The present study clearly has some limitations. The most important limitation is that the framework remains conceptual and has not yet been fully implemented or empirically tested as an integrated system, and requires further validation. Further limitations relate to the reliability of patient-entered data. PGHD has been identified as a means of extending data collection beyond clinical environments, but there is still considerable uncertainty with regard to accuracy, privacy, interoperability, and workflow integration [5, 11]. As patients may provide incomplete or inaccurate records, and to the need for strong privacy, security, and data-governance mechanisms. Mars and Scott (2022) note that the perceived benefits exceed the available empirical evidence for improved collaborative care, but it has not yet been established how heterogeneous patient-entered data should be validated, prioritised, and

converted into actionable information [18]. In addition, the effectiveness of the knowledge base depends on the accuracy, and maintaining accurate personalised rules, clinical targets, and guideline-based knowledge is difficult, since incorrect or outdated information could affect how future outputs are interpreted.

These limitations suggest that future work is needed and must concentrate on expert review for validation, system refinement, pilot implementation, and evaluation using real-world clinical and patient-generated data. Further research is needed to establish whether the framework is usable, interpretable, safe, and acceptable to patients, clinicians, and healthcare organisations. From a policy perspective, the framework highlights the need for clear governance around data security, consent, clinical accountability, and the use of patient-generated health data in care pathways. For intervention design, the proposed framework may serve as a useful basis for developing CKD self-management systems that improve clinical visibility between consultations, support personalised feedback, and maintain human oversight. Overall, this study provides a foundation for future development and evaluation of a clinically accountable and patient-centred Chronic Kidney Disease self-management system.

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